



Semester 1 Examinations 2016 / 2017

Exam Code(s)	1SD1/1SD3/1MF1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development.
Module Code(s) Module(s)	CT874 Programming I
Paper No.	I
External Examiner(s) Internal Examiner(s)	Professor Liam Maguire Professor Peter McHugh Dr. Jim Duggan * Seamus Hill
<u>Instructions:</u>	Candidates are required to answer: Section A (20 Marks) Answer all 10 parts of section A. This section consists of 10 multiple-choice questions, all of which should be attempted. Answers are to be written on the MCQ sheet provided, NOT on the Examination paper. Section B (80 Marks) Answer Any Two Questions. All questions carry equal marks.
Duration No. of Pages Department(s)	2 hours 7 Information Technology
Requirements	MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

Q1.

- 1) Consider the following code:

```
public class X {  
    public void parentMethod(){  
        System.out.println("parent");  
    }  
}  
  
class Y extends X {  
    public static void main(String args[]){  
        Y y = new Y();  
        y.parentMethod();  
    }  
  
    public static void myMethod(){  
        System.out.println("child");  
    }  
}
```

What is the output of this code?

- a) parent
 - b) child
 - c) X
 - d) There is no output
- 2) Given the following code fragment, what is the value of z after of the following?

```
static Object boxes[] = new Object[10];  
boxes[0] = 10;  
int z = boxes[0];
```

- a) 10
- b) 0
- c) code failed to compile
- d) Object

- 3) What is the output of the following code:

```
String a = "cigar";  
  
for(int i=1; i<a.length(); i+=2){  
    System.out.print(a.charAt(i));  
}
```

- a) The output will be "igar"
- b) The output will be cigar
- c) The output will be cgr
- d) An exception will be thrown

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4) What is the output of this code:

```
public class Array {  
    public static void main(String[] args){  
  
        float[] a = new float[3];  
        float x = 0, y = x;  
  
        for(int i=0; i<a.length; i++){  
            x += a[i];  
        }  
        System.out.print(y += x/a.length);  
    }  
}
```

- a) An exception will be thrown
- b) 3.0
- c) 0.0
- d) There is no output

5) Examine the following code:

```
class Bus extends Vehicle {  
  
    public Bus(){  
        super();  
    }  
  
}
```

What does super do?

- a) Makes a call to the Object() constructor.
- b) Makes a call to the Bus() constructor.
- c) Makes a call to the Vehicle() constructor.
- d) Makes a call to the superclass of Vehicle's constructor.

6) Complete the following sentence: A(n) _____ has a method prototype but no body.

- a) default constructor
- b) default method
- c) concrete method
- d) abstract method

7) The practice of writing multiple constructors to handle different sets of inputs is known as _____.

- a) Inheritance
- b) Polyinputting
- c) overloading
- d) multiplicity

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8) What is the problem with the following code:

```
public class Bag {  
    private String colour;  
  
    private Bag(){  
        colour = "unknown";  
    }  
  
    public String getColour(){  
        return colour;  
    }  
  
    class New {  
        Bag b;  
  
        public void getBag(){  
            return b;  
        }  
    }  
}
```

- a) The accessor method needs a different return value.
- b) b has not yet been initialized.
- c) The class needs more methods.
- d) The method getBag needs more statements.

9) What is the output of the following code?

```
public class TryCatch {  
  
    public static void main(String[] args){  
  
        try{  
            int number1 = Integer.parseInt("I542");  
            System.out.print("number1 is ok ");  
        }catch(NumberFormatException e){  
            System.out.print("Error");  
        }  
    }  
}
```

- a) number1 is ok
- b) There is no output
- c) Error
- d) number1 is ok Error

10) What is meant by “returning an object” from a method?

- a) It means you are returning a null value to the caller.
- b) It means returning a primitive type to the caller.
- c) It means returning the object’s address to the caller.
- d) It means returning the object’s value to the caller.

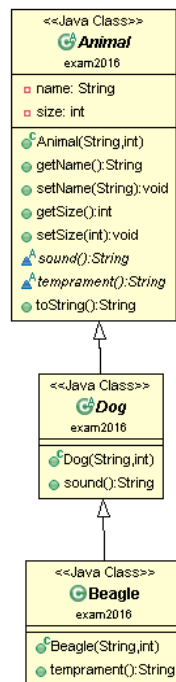
Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) In relation to the *Serializable* interface, detail your understanding of the four steps involved in serializing an object, use code fragments where appropriate.

[8 Marks]

- b) Using the following UML diagram you are required to do the following:



- i. Using Java, write code which implement the **Animal**, **Dog** and **Beagle** classes. Note: The **Animal** class implements the *Serializable* interface.

[15 Marks]

- ii. Write a tester class, which creates an instance of the **Beagle** class and set the instance variables. The tester class should **serialize** and **deserialize** the object.

[17 Marks]

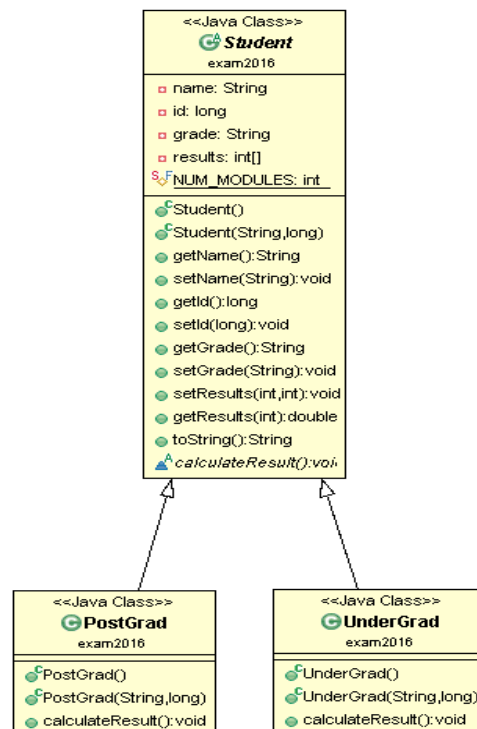
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Q 3.

- a) Outline in your own words the meaning of the terms **abstract class** and **abstract method** as used in object-oriented programming.

[6 Marks]

- b) ACME University requires you to write an application in Java. The exam rules within the university state that to pass the semester, a PostGrad student requires an average of **50** or above for the 5 modules taken. An underGrad student, on the other hand, requires an average of **40** or above to pass the 5 modules taken.



- i. You are required to write the *Student*, *PostGrad* and *UnderGrad* classes outlined in the UML diagram above and the application is to be **Serializable**. Rules relating to passing the semester are to be implemented in the respective *calculateResults()* methods.

[12 Marks]

- ii. Write a tester class, which creates instances of the *UnderGrad* and *Postgrad* classes and puts them into an **array** of type *Student*. The tester class should use a loop to take each student's marks as input from the user. The tester should then calculate each students result and display whether or not they have passed the semester.

[12 Marks]

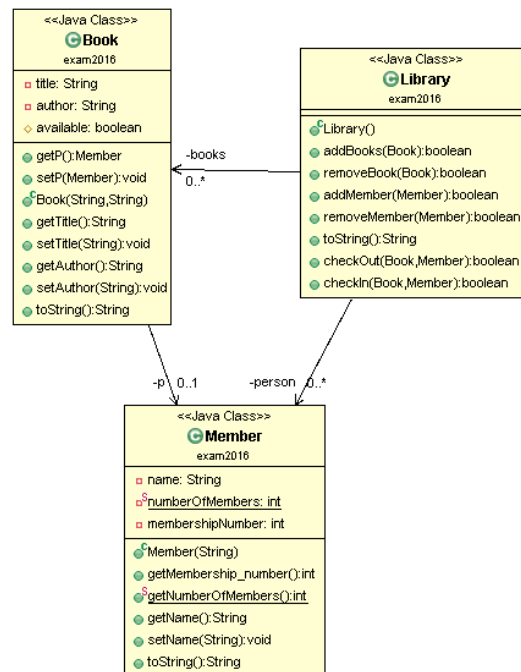
- iii. Add code that allows the tester class to *serialize* your application.

[10 Marks]

PTO

Q 4.

The local library is looking for a Java application to keep track of its stock of Books and Members. Using the following UML diagram as a guide, you are required to:



- Create the Library class with the following data members:
private ArrayList<Book> **books**;
private ArrayList<Member> **members**;
 [8 Marks]
- Write the Member Class which contains the following data member:
private String **name**;
private static int **numberOfMember**;
private int **membershipNumber**;
 [4 Marks]
- Implement the Book class with the following data members:
private String **title**;
private String **author**;
protected boolean **available**;
private Member **member** = **null**;
 [6 Marks]
- Write a tester class and implement the following:
 - Create an instance of the *Library* class, two instances of the *Member* class and two instances of the *Book* class. Set the values of the data members and add the *Member* and *Book* instances should to the respective collections (*members* & *books*) and display the contents of the collections.
 [10 Marks]
 - Check out a book and display its availability after being checked out.
 [6 Marks]
 - Check in a book and display its availability after being checked in.
 [6 Marks]